Simulations of the gravitational lensing of star fields by black holes, similar to those that underlie *Interstellar*, have been carried out previously by a number of physicists and can be found on the web. Especially impressive are those by Alain Riazuelo; see www2.iap.fr/users/riazuelo/interstellar. See also the section on Chapter 28, below.

Paul Franklin's team and I plan to write several somewhat technical articles about the simulations that they carried out using the equations I gave them: the simulations underlying *Interstellar*'s images of Gargantua and its disk and the wormhole, and additional simulations that have revealed surprising things. You can access these articles on the web at http://arxiv.org/find/gr-qc.

Chapter 9. Disks and Jets

For in-depth discussions of quasars, accretion disks, and jets, see *Gravity's Fatal Attraction* (Begelman and Rees 2009), Chapter 9 of *Black Holes & Time Warps* (Thorne 1994), and at a more technical and more detailed level, *Black Hole Astrophysics* (Meier 2012). For the tidal disruption of stars by black holes and the resulting accretion disks, see the website of James Guillochon (who, with colleagues, was responsible for the simulations that underlie Figures 9.5 and 9.6): http://astrocrash.net/projects/tidal-disruption-of-stars/. For astrophysically realistic film clips of accretion disks and their jets, I recommend some by Ralf Kaehler (Stanford University) at http://www.slac.stanford.edu/~kaehler/homepage/visual izations/black-holes.html, based on simulations by Jonathan C. McKinney, Alexander Tchekhovskoy, and Roger D. Blandford (McKinney, Tchekhovskoy, and Blandford 2012). For some images of accretion disks with Doppler shifts included as well as gravitational lensing, see the website of the